# Quiz 1

15 minutes · closed book · no calculators

Three questions, five minutes each. Drawn from the core problems of Homeworks 1 and 2. Answers should be short; a correct one-line argument is worth more than a page of work.

**1.** COMPUTE Let $X \sim \mathcal{N}(0, \sigma^2)$ on $\mathbb{R}$.

(a) Write down the differential entropy $h(X)$.

(b) A colleague reports that a certain real-valued random variable has differential entropy $-3$. Is this possible? Give an example or explain why not.

**2.** COMPUTE Observations follow $y = Hw + \sigma\xi$, with $\xi \sim \mathcal{N}(0, I_n)$ and a prior $w \sim \mathcal{N}(0, \tau^2 I_p)$ independent of $\xi$.

(a) Write down $\mathbb{E}[w \mid y]$.

(b) Identify the regularization parameter in terms of $\sigma$ and $\tau$, and say in one sentence what a large value of it means.

**3.** WHICH HYPOTHESIS FAILS Let $g \sim \mathcal{N}(0, 1)$ and let $\varepsilon$ be an independent random sign. Set

$$X = g, \qquad Y = \varepsilon g.$$

(a) Show $\mathrm{Cov}(X, Y) = 0$.

(b) Are $X$ and $Y$ independent? Justify in one sentence.

(c) Both $X$ and $Y$ are standard normal and they are uncorrelated. Name precisely the hypothesis of the theorem from Lecture 1 that fails here.

Name: ______________________________